



5 (a) State Kirchhoff's second law.

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 [1]

(b) A battery of electromotive force (e.m.f.) 9.0V and negligible internal resistance is connected in series with a variable resistor X and a thermistor Y as shown in Fig. 5.1.

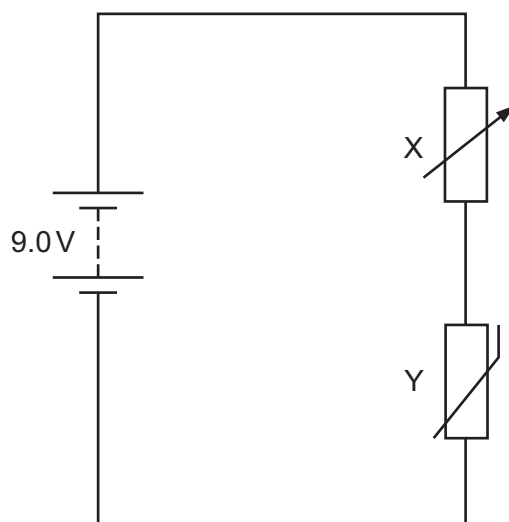


Fig. 5.1

Fig. 5.2 shows the relationship between temperature and resistance for the thermistor.

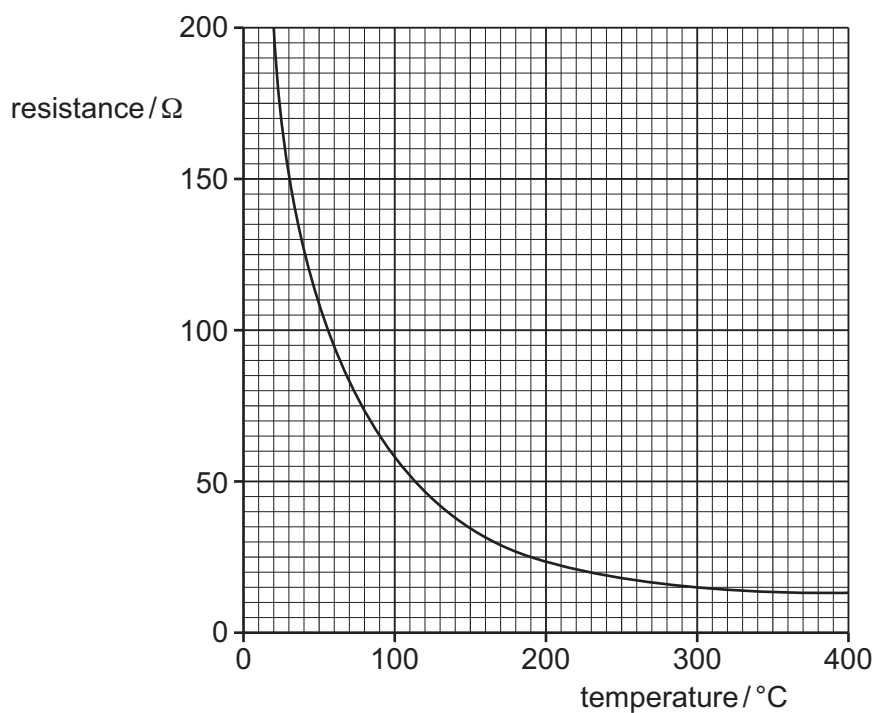


Fig. 5.2





- (i) The current in the circuit is $1.1 \times 10^{-2} \text{ A}$. The potential difference across Y is 4.0 V.

Calculate the resistance of X.

resistance = Ω [2]

- (ii) The temperature of Y is changed to 190°C . The resistance of X remains unchanged.

Determine the new potential difference across Y.

potential difference = V [3]

- (iii) The resistance of X is increased. The temperature of Y remains at 190°C .

By reference to the current in the circuit, state and explain the effect of this change, if any, on the potential difference **across Y**.

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..... [3]

[Total: 9]

